

AI-DRIVE VOICE-ACTIVATED WHEELCHAIR WITH REAL-TIME LOCATION MONITORING

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

ELECTRICAL AND ELECTRONICS ENGINEERING



ANNA UNIVERSITY REGIONAL CAMPUS COIMBATORE – 641046

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BONAFIDE CERTIFICATE

Certified that this project report “**AI-DRIVE VOICE-ACTIVATED WHEELCHAIR WITH REAL-TIME LOCATION MONITORING**” is the Bonafide work of **PRIYADHARSHAN A (710022105019), SRIVATHSHA KALEESHWARAN D (710022105011), SHANTHINI V (710022105702)** who carried out the project work under my supervision.

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ABSTARCT

The project titled “**AI-Powered Voice-Activated Wheelchair with Real-Time Location Monitoring**” is designed to assist individuals with mobility impairments by providing an intelligent, safe, and user-friendly transportation system. The system enables users to control a wheelchair using voice commands, reducing the need for physical effort and improving independence in daily movement.

The wheelchair integrates voice recognition technology with an embedded control system to interpret user commands such as move forward, backward, left, right, and stop. These commands are processed using a microcontroller, which controls the motor driver to navigate the wheelchair accordingly. In addition, the system incorporates a GPS-based tracking module to enable real-time location monitoring.

The **DHRUV-PRO NEO-6M GPS Module** is used to capture accurate latitude and longitude data through satellite communication. This information is processed using a microcontroller such as the **NodeMCU ESP8266**, which transmits the location data to a cloud server via Wi-Fi connectivity. The live location is then displayed on a web interface using mapping services like the **Google Maps API**, allowing caregivers or family members to monitor the user in real time.

The system improves safety by enabling continuous tracking and emergency monitoring while ensuring ease of mobility through voice-based control. This integration of artificial intelligence, embedded systems, and IoT technologies makes the solution efficient, cost-effective, and suitable for real-world assistive applications.

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LIST OF ABBREVIATIONS

NLP	Natural Language Processing
BCI	Brain Computer Interface
IoT	Internet of Things
GPS	Global Positioning System
GSM	Global System for Mobile Communication
Wi-Fi	Wireless Fidelity
RF	Radio Frequency
ASR	ASR – Automatic Speech Recognition
TTS	Text To Speech
NLU	Natural Language Understanding
IR – Infrared	IR – Infrared
US	Ultrasonic Sensor
UART	Universal Asynchronous Receiver Transmitter
ADC	Analog to Digital Converter
ADC	ADC – Analog to Digital Converter
DAC	Digital to Analog Converter
PWM	Pulse Width Modulation

CHAPTER 1

INTRODUCTION

Mobility impairment is a serious condition that affects a large number of people across the world, making it difficult for them to move independently in daily life. Most users depend on caregivers or manual wheelchairs, which can reduce their freedom and may not always ensure safety, especially during emergencies or outdoor movement. To solve this problem, modern technologies such as Artificial Intelligence (AI), IoT, and embedded systems are being used to create smart assistive devices. The project “**AI-Powered Voice-Activated Wheelchair with Real-Time Location Monitoring**” is developed to provide an intelligent and user-friendly mobility solution that allows individuals with physical disabilities to control a wheelchair using voice commands. These commands, such as move forward, backward, left, right, and stop, are captured through a voice recognition system and processed by a microcontroller, which converts them into electrical signals to control the motors of the wheelchair smoothly and accurately.

In addition to movement control, the system also focuses on user safety through real-time location tracking. A **DHRUV-PRO NEO-6M GPS Module** is used to receive signals from satellites and determine the exact position of the wheelchair using the concept of **Trilateration**, which calculates location based on distance from multiple satellites. The collected GPS data, including latitude and longitude, is processed using a microcontroller such as **NodeMCU ESP8266**, which then sends the information to a cloud server via WiFi connectivity.

This live data is displayed on a web interface using **Google Maps API**, allowing caregivers or family members to monitor the user’s exact location in real time. On a global scale, developed countries like the USA, Japan, and Germany have

already introduced advanced smart wheelchairs with features such as AI navigation, obstacle detection, and brain-computer interfaces, but these systems are very expensive and not widely accessible. In comparison, developing countries still depend on basic wheelchairs due to cost and infrastructure limitations. Therefore, this project provides a low-cost, efficient, and practical solution by combining voice control and real-time tracking, ultimately improving independence, safety, and mobility for users with disabilities.

1.1 WHEELCHAIR APPLICATIONS

The **AI-Powered Voice-Activated Wheelchair with Real-Time Location Monitoring** has a wide range of practical applications in healthcare, assistive technology, and smart mobility systems. It is mainly designed to improve independence, safety, and quality of life for individuals with mobility impairments.

This system is widely used in **home care environments**, where users can move independently without constant caregiver assistance. It is also highly useful in **hospitals and rehabilitation centers**, where patients recovering from injuries or surgeries need safe and controlled mobility support. In **elderly care facilities**, the system helps senior citizens move comfortably using simple voice commands, reducing physical strain and dependency on others.

Another important application is in **outdoor navigation and public spaces**, where real-time GPS tracking ensures safety by allowing caregivers to monitor the user's location continuously. Using the **DHRUV-PRO NEO-6M GPS Module**, the system provides accurate location updates, which are displayed through web platforms using **Google Maps API**, making it useful for emergency tracking and rescue situations.

In addition, the system can be applied in **smart healthcare systems**, where AI and IoT technologies are used to develop intelligent assistive devices. It also supports **emergency response scenarios**, helping locate users quickly in case of accidents or distress. Overall, this wheelchair system plays a key role in modern assistive technology by improving accessibility, safety, and independent mobility for physically challenged individuals.

1.2 LITERATURE SURVEY

1.2.1 Voice-Controlled Smart Wheelchair with Tracking

Alkhalid, A and Oleiwi, B (2019) explained that the paper presents a smart wheelchair system controlled through voice commands with an integrated tracking mechanism. The proposed system is designed to support physically disabled individuals by improving mobility and independence. The wheelchair uses speech recognition technology to receive user commands such as forward, backward, left, right, and stop. The system also includes a tracking feature that helps monitor the wheelchair location for safety and emergency purposes. The authors focused on improving user comfort, navigation accuracy, and communication efficiency. Their work contributes to the development of intelligent assistive technologies for healthcare and rehabilitation applications.

1.2.2 Voice Controlled Wheelchair Using Esp32

Anil, A.R, Devaraj., M.P, Guruprasad., S.S and Kumaraswamy, D (2026) explained that the proposed system utilizes the ESP32 microcontroller for implementing a voice-controlled wheelchair. The ESP32 enables wireless communication, real-time processing, and low power consumption. The wheelchair responds to predefined voice instructions and controls motor movements accordingly. The authors integrated Bluetooth technology for communication

between the mobile application and the wheelchair system. The proposed design improves accessibility for disabled users and reduces the need for manual wheelchair operation. The system offers cost-effective implementation with efficient performance for smart mobility applications.

1.2.3 Simulation of Autonomous Robotic Vehicle Navigation

Bakirci, M (2023) explained that the study focuses on autonomous robotic vehicle navigation through simulation-based analysis. The paper discusses navigation algorithms, obstacle avoidance methods, and path planning strategies for autonomous systems. The proposed simulation environment evaluates the movement accuracy and decision-making capability of robotic vehicles under different operating conditions. The research highlights the importance of sensor integration and intelligent control methods for safe autonomous navigation. This work contributes to the development of autonomous wheelchair navigation and robotic mobility systems.

1.2.4 Voice Controlled Wheelchair

Bharade, A., Kumar, R., Mulge., S and Huded, R (2025) explained that the paper presents a wheelchair controlled through speech recognition technology for assisting physically challenged individuals. The system converts voice commands into movement instructions using embedded controllers and motor driver circuits. The proposed wheelchair enhances user independence and minimizes physical effort during operation. The authors focused on achieving reliable voice recognition and smooth directional control. The system also improves accessibility and provides an affordable solution for smart assistive mobility devices.

1.2.5 Speech Recognition in Noisy Environments

Cai, T.F et al. (2022) explained that the study investigates speech recognition performance in noisy environments. The paper discusses advanced noise reduction techniques, signal processing methods, and machine learning algorithms for improving speech recognition accuracy. The proposed methods help systems identify voice commands even in industrial or crowded surroundings. The research highlights the challenges faced in real-time speech processing and proposes efficient solutions for robust communication. This work is highly useful for developing reliable voice-controlled wheelchairs in practical environments.

1.2.6 Intelligent Control of Power Wheelchairs

Cooper, R.A (1995) explained that the paper introduces intelligent control strategies for powered wheelchairs to improve mobility assistance for disabled users. The study discusses navigation support, obstacle avoidance, and adaptive control systems for safe wheelchair operation. The proposed intelligent wheelchair system enhances user safety and reduces operational difficulties. The authors evaluated different control mechanisms to achieve better manoeuvrability and driving performance. Their work laid the foundation for future research in smart wheelchair technologies and autonomous assistive systems.

1.2.7 Embedded Voice Recognition System for IOT Devices

Coucke, A et al. (2018) explained that the paper presents an embedded voice recognition system designed for IoT devices with low computational requirements. The proposed system processes voice commands locally without relying heavily on cloud services. The study focuses on lightweight machine learning models suitable for embedded hardware platforms. The authors demonstrated efficient voice recognition performance with reduced latency and power consumption. This

technology is beneficial for implementing smart wheelchairs using embedded voice-controlled systems.

1.2.8 Data Augmentation for Speech Command Recognition

De Simone, G et al. (2023) explained that the paper focuses on improving speech command recognition accuracy through data augmentation techniques. The proposed methods increase the diversity of speech datasets by adding variations such as background noise, pitch changes, and time shifts. The study improves machine learning model performance for recognizing commands under different environmental conditions. The research contributes to robust voice recognition systems for intelligent mobility and assistive technologies.

1.2.9 Smart Wheelchair with AI-Based Navigation

Ferracuti, F et al. (2025) explained that the study presents a smart wheelchair integrated with artificial intelligence-based navigation systems. The wheelchair uses AI algorithms for path planning, obstacle detection, and autonomous movement. The proposed system improves navigation efficiency and ensures user safety in indoor and outdoor environments. The authors utilized intelligent sensors and real-time decision-making techniques for autonomous mobility assistance. Their work significantly contributes to advanced healthcare mobility systems.

1.2.10 Bluetooth Low Energy for IOT Systems

Gomez, S., Oller, J and Paradells, J (2012) explained that the paper discusses Bluetooth Low Energy (BLE) technology for IoT communication systems. BLE provides low power consumption, wireless connectivity, and efficient data

transmission between embedded devices. The study highlights the suitability of BLE for healthcare monitoring and assistive technologies. The proposed communication method is useful for connecting smartphones and embedded controllers in voice-controlled wheelchair systems.

1.2.11 Voice Recognition Based Intelligent Wheelchair and GPS Tracking System

Jalajakshi, Y., Amura, B.T., Naveena, B., Reddy and Ramanjaneyulu, R (2025) explained that the paper presents an intelligent wheelchair system integrated with voice recognition and GPS tracking technology. The proposed system enables disabled individuals to control wheelchair movement through speech commands. GPS tracking is incorporated to monitor the user's real-time location and improve safety during emergency situations. The authors focused on enhancing mobility assistance, navigation accuracy, and user independence. The system combines voice processing modules, embedded controllers, and wireless communication techniques for efficient wheelchair operation.

1.2.12 ESP 32 Microcontroller in Embedded Systems

Kareem, H and Dunaev, D (2021) explained that the study discusses the application of the ESP32 microcontroller in modern embedded systems. ESP32 provides high processing capability, Wi-Fi and Bluetooth connectivity, low power consumption, and efficient real-time control functions. The paper highlights the advantages of ESP32 for IoT-based smart applications and automation systems. The authors demonstrated that ESP32 supports reliable communication and flexible hardware interfacing. This technology is highly suitable for implementing voice-controlled wheelchair systems and intelligent healthcare devices.

1.2.13 A Comprehensive Review of Smart Wheelchairs

Leaman, J and La, H.M (2017) explained that the paper provides a comprehensive review of smart wheelchair technologies and assistive mobility systems. The study discusses various control methods including voice control, joystick operation, brain-computer interfaces, and autonomous navigation techniques. The authors analyzed obstacle avoidance systems, sensor integration, and intelligent control algorithms used in modern wheelchairs. The paper also highlights challenges related to safety, usability, and system reliability. Their review contributes valuable insights for the development of advanced intelligent wheelchair technologies.

1.2.14 ESP 32 for IOT Applications

Maier, A., Sharp, A and Vagapov, Y (2017) explained that the paper presents the implementation of ESP32 microcontrollers for Internet of Things (IoT) applications. The ESP32 supports wireless communication, sensor interfacing, and real-time data processing with low energy consumption. The authors discussed the hardware architecture and software capabilities of the ESP32 platform for smart embedded systems. The paper demonstrates the effectiveness of ESP32 in automation and wireless control applications. This work supports the development of intelligent wheelchairs with Bluetooth and Wi-Fi-based communication systems.

1.2.15 Smart Wheelchair with Voice Control and Environmental Monitoring

Munasinghe, K.K.T (2025) explained that the study introduces a smart wheelchair system integrated with voice control and environmental monitoring technologies. The wheelchair responds to speech commands and simultaneously

monitors environmental conditions using sensors. The proposed system improves user safety by detecting obstacles and monitoring surrounding parameters. The authors focused on enhancing mobility support, intelligent navigation, and real-time environmental awareness. Their work contributes to the development of advanced assistive healthcare systems for physically disabled individuals.

1.2.16 Smart Assistive Device with Voice Guidance

Olayiwola., A et al. (2025) explained that the paper presents a smart assistive device equipped with voice guidance technology for disabled users. The system provides voice-based interaction and navigation support for improving user accessibility and communication. The proposed device integrates sensors, embedded processors, and intelligent speech systems for real-time assistance. The study focuses on enhancing independent mobility and reducing operational complexity. The authors demonstrated that voice guidance systems improve the effectiveness of healthcare assistive devices.

1.2.17 Development of an Autonomous Driving Smart Wheelchair

Ryu, S et al. (2021) explained that the paper focuses on the development of an autonomous driving smart wheelchair system for physically challenged individuals. The wheelchair uses intelligent navigation algorithms, obstacle detection sensors, and autonomous driving mechanisms. The proposed system enables safe and efficient movement in complex environments without continuous manual control. The authors implemented real-time decision-making methods and path planning techniques to improve wheelchair navigation performance. Their work contributes significantly to autonomous healthcare mobility systems.

1.2.18 Speech Recognition for Industrial Environments

Saggese, B et al. (2024) explained that the study investigates speech recognition technologies designed for industrial and noisy environments. The paper discusses noise filtering methods, signal enhancement techniques, and robust speech processing algorithms. The proposed approaches improve command recognition accuracy under challenging environmental conditions. The authors focused on developing reliable voice recognition systems capable of operating in real-time applications. This work is beneficial for improving the performance of voice-controlled wheelchairs in practical environments.

1.2.19 Speech Recognition-Based Wireless Control System for Robotics

S.G et al. (2025) explained that the paper presents a wireless robotic control system based on speech recognition technology. The system converts user voice commands into wireless control signals for robotic movement and operation. The proposed design integrates speech processing modules, wireless communication systems, and embedded controllers for efficient performance. The authors focused on improving response accuracy, communication reliability, and real-time control capabilities. This work supports the development of intelligent voice-controlled wheelchairs and robotic healthcare applications.

1.3 PROBLEM STATEMENT

The increasing number of individuals suffering from mobility impairments has created a significant demand for intelligent assistive systems that can enhance independent movement, safety, and real-time monitoring, however existing wheelchair technologies still present major limitations in terms of accessibility,

affordability, and integration, as most traditional systems rely on manual propulsion or joystick-based control mechanisms which require physical effort and are not suitable for users with severe disabilities, while advanced powered wheelchairs, although more convenient, still lack intelligent decision-making capabilities and do not provide sufficient autonomy or safety features in dynamic environments, furthermore many existing research prototypes in smart wheelchair systems focus on isolated functionalities such as voice recognition, obstacle detection, or GPS tracking, but fail to integrate these components into a single unified framework, thereby reducing their practical usability in real-world scenarios, additionally voice-controlled wheelchair systems, although widely researched, suffer from several limitations including poor recognition accuracy in noisy environments, dependency on predefined command sets, latency in response time, and lack of multilingual support, which restrict their effectiveness for diverse user populations, similarly sensor-based navigation systems that utilize ultrasonic, infrared, or LiDAR sensors provide obstacle detection capabilities but often lack contextual decision-making intelligence and require high computational resources, making them unsuitable for low-cost implementations, in parallel GPS-based tracking systems have been introduced in assistive devices to enhance user safety by enabling location monitoring, where modules such as the **DHRUV-PRO NEO-6M GPS Module** are commonly used due to their affordability and ability to provide accurate positioning through satellite communication using the principle of **Trilateration**, however these systems often suffer from signal degradation in indoor environments and lack seamless integration with real-time visualization platforms, while IoT-based systems using microcontrollers such as **NodeMCU ESP8266** have improved connectivity by enabling wireless data transmission to cloud servers, they still depend heavily on stable internet connectivity and require additional optimization for real-time performance, moreover modern visualization techniques using **Google Maps API** allow real-time tracking of wheelchair users, but latency issues and server

dependency remain challenges in critical applications, further advancements in AI-based wheelchair systems have introduced autonomous navigation using machine learning, computer vision, and sensor fusion techniques, enabling obstacle avoidance and path planning, yet these systems are computationally expensive and not suitable for low-resource environments, in addition alternative control systems such as eye-tracking, gesture recognition, and brain-computer interfaces (BCI) have been explored for users with severe disabilities, but these technologies require specialized hardware and complex calibration processes, making them impractical for widespread adoption, therefore despite significant advancements in smart wheelchair research globally, there remains a clear gap in developing a low-cost, fully integrated, and user-friendly assistive mobility system that combines voice control for intuitive operation, GPS-based real-time tracking for safety monitoring, and IoT connectivity for remote access, hence the proposed system aims to address these challenges by integrating AI-powered voice recognition, embedded microcontroller-based motor control, and real-time GPS tracking into a single unified architecture, thereby improving mobility independence, enhancing user safety, reducing caregiver dependency, and providing a scalable solution suitable for both urban and rural healthcare environments.

1.4 OBJECTIVE OF THE PROJECT

The main objective of the project **“AI-Powered Voice-Activated Wheelchair with Real-Time Location Monitoring”** is to design and develop an intelligent assistive mobility system that enables users with mobility impairments to move independently, safely, and efficiently using voice commands. The system aims to eliminate the dependency on manual wheelchair control or caregiver assistance by providing a hands-free operation mechanism through AI-based voice recognition, ensuring ease of use for physically challenged individuals.

Another key objective is to implement a reliable **voice-controlled navigation system** that accurately interprets user commands such as forward, backward, left, right, and stop, and converts them into real-time motor control signals for smooth wheelchair movement. This ensures intuitive interaction between the user and the system, improving accessibility and usability.

A major objective of the project is to integrate **real-time location tracking** using the **DHRUV-PRO NEO-6M GPS Module**, which collects geographic coordinates using satellite signals and the principle of **Trilateration**. The collected data is processed and transmitted using a microcontroller such as **NodeMCU ESP8266**, enabling continuous monitoring of the user's location over the internet.

The system also aims to develop an **IoT-based monitoring platform** where real-time GPS data is sent to a cloud server and displayed using web-based mapping tools such as the **Google Maps API**, allowing caregivers or family members to track the user's movement and respond quickly in case of emergencies.

Another important objective is to enhance user safety by enabling continuous tracking and monitoring, reducing the risk of getting lost or facing emergencies without assistance. The system is also designed to be cost-effective and efficient so that it can be easily implemented in real-world applications, especially in developing regions where advanced assistive technologies are often unaffordable.

Finally, the project aims to integrate multiple technologies such as AI, embedded systems, IoT, and GPS into a single unified framework to create a smart, scalable, and practical mobility solution that improves independence, safety, and quality of life for individuals with mobility impairments.

1.5 MOTIVATION AND SCOPE OF THE PROJECT

The primary motivation behind this project arises from the increasing need to support individuals with mobility impairments who face significant difficulties in performing daily activities independently. Many users rely on manual wheelchairs or caregiver assistance, which reduces their freedom, confidence, and ability to move safely in both indoor and outdoor environments. In emergency situations, the lack of real-time monitoring further increases safety risks. Therefore, there is a strong need for an intelligent, affordable, and user-friendly mobility solution that can improve independence and ensure continuous safety monitoring. This need motivates the development of an AI-powered system that combines voice control and real-time tracking into a single integrated platform.

Another important motivation is the rapid advancement of technologies such as Artificial Intelligence, Internet of Things (IoT), and embedded systems, which have made it possible to design smart assistive devices at low cost. Voice recognition technology enables hands-free control, which is especially useful for users who cannot operate traditional joystick-based systems. Similarly, GPS-based tracking enhances safety by allowing caregivers to monitor the user's location in real time, reducing risks during travel or outdoor movement. The integration of these technologies aims to bridge the gap between high-end expensive systems and affordable real-world solutions.

The scope of this project is wide and extends across multiple domains including healthcare, assistive robotics, IoT-based monitoring systems, and smart mobility solutions. The system utilizes the **DHRUV-PRO NEO-6M GPS Module** for accurate location tracking, which works based on satellite communication and the principle of **Trilateration**. The data is processed using microcontrollers such as **NodeMCU ESP8266**, which enables wireless communication and real-time data transmission to cloud platforms. The live location is then visualized using web

technologies like the **Google Maps API**, making remote monitoring simple and effective.

From an application perspective, the system can be used in hospitals, rehabilitation centers, elderly care homes, and home-based care environments to improve mobility assistance. It also has potential use in emergency response systems where quick location tracking is critical. In the future, the system can be expanded by integrating advanced features such as obstacle detection, AI-based path planning, mobile app control, and voice assistant integration.

Overall, the project scope is not limited to wheelchair control alone but extends toward developing a complete smart assistive mobility ecosystem that enhances independence, safety, and quality of life for users with physical disabilities while remaining cost-effective and scalable for real-world deployment.

1.6 THESIS ORGANIZATION

This thesis is systematically organized to present the complete design, development, implementation, and evaluation of the AI-Powered Voice-Activated Wheelchair with Real-Time Location Monitoring system in a structured and logical manner, beginning with the introductory chapter that provides a comprehensive overview of mobility impairment challenges faced by individuals globally, the limitations of existing manual and semi-automated wheelchair.

This thesis is systematically organized to present the complete design, development, implementation, and evaluation of the AI-Powered Voice-Activated Wheelchair with Real-Time Location Monitoring system in a structured and logical manner, beginning with the introductory chapter that provides a comprehensive overview of mobility impairment challenges faced by individuals globally, the

limitations of existing manual and semi-automated wheelchair systems, and the growing need for intelligent assistive technologies that integrate artificial intelligence, embedded systems, and IoT-based monitoring to improve independence and safety, followed by a detailed literature survey chapter that critically analyses previous research works in the domain of smart wheelchairs, including voice-controlled systems, sensor-based navigation techniques, GPS tracking implementations, IoT-based monitoring frameworks, and advanced AI-driven autonomous mobility solutions, while identifying key limitations such as high cost, limited accuracy in noisy environments, lack of real-time integration, and absence of unified system architectures, and then proceeding to the problem statement, objectives, and scope chapter which clearly defines the real-world challenges faced by users with mobility impairments, highlights the necessity for a low-cost and efficient solution, and establishes the goals of integrating voice recognition, GPS tracking, and IoT communication into a single system, after which the system design and architecture chapter elaborates on the hardware and software framework of the proposed model, including the use of embedded microcontrollers, wireless communication modules, motor driver circuits, and GPS technology such as the **DHRUV-PRO NEO-6M GPS Module**, which operates based on satellite signal reception and the principle of **Trilateration** for accurate position calculation, along with the use of WiFi-enabled controllers like **NodeMCU ESP8266** for real-time data transmission, and web-based visualization tools such as **Google Maps API** for live tracking display, followed by the implementation chapter which provides detailed explanations of hardware integration, circuit connections, programming logic, voice command processing, GPS data acquisition, and cloud-based communication mechanisms, ensuring seamless interaction between all system components, and then the results and discussion chapter which evaluates system performance in terms of accuracy of voice recognition, reliability of GPS tracking, response time of motor control, and overall system efficiency under different

operating conditions, highlighting both advantages and limitations observed during testing, and finally concluding with the conclusion and future enhancement chapter which summarizes the outcomes of the project and suggests potential improvements such as integration of advanced AI-based obstacle detection systems, mobile application control interfaces, enhanced voice processing algorithms, and autonomous navigation capabilities, thereby establishing a foundation for future research in intelligent assistive mobility systems aimed at improving accessibility, safety, and independence for individuals with mobility impairments in real-world environments.

CHAPTER 2

DESIGN AND METHODOLOGY AI-DRIVE VOICE-ACTIVATED WHEELCHAIR WITH REAL-TIME LOCATION MONITORING

The design and methodology of the **AI-Driven Voice-Activated Wheelchair with Real-Time Location Monitoring** focuses on developing an integrated assistive mobility system that enables users with physical disabilities to control movement through voice commands while ensuring continuous safety monitoring through real-time tracking. The system architecture is designed with three main units: input, processing, and output. In the input unit, voice commands such as forward, backward, left, right, and stop are captured through a microphone-based voice recognition system, which converts human speech into digital signals using AI-based speech processing techniques.

These commands are then transmitted to the processing unit, where a microcontroller such as **NodeMCU ESP8266** acts as the central controller to interpret instructions and manage system operations. Based on the received commands, the microcontroller sends signals to the motor driver circuit, which controls the DC motors attached to the wheelchair, enabling smooth and accurate movement.

Alongside mobility control, the system incorporates a GPS-based tracking module such as the **DHRUV-PRO NEO-6M GPS Module**, which continuously receives satellite signals and determines precise location using the principle of **Trilateration**. The collected location data is processed and transmitted via WiFi to a cloud server using the NodeMCU module, allowing real-time updates. The live location is then displayed on a web interface using **Google Maps API**, enabling

caregivers or family members to monitor the user's movement continuously for safety purposes. The methodology ensures that all components work in synchronization, where voice control handles mobility, GPS ensures tracking, and IoT enables communication between the device and remote users. Overall, this integrated design provides a low-cost, efficient, and intelligent solution for improving independence, safety, and real-time monitoring of individuals with mobility impairments.

2.1 THEORITICAL SYSTEM ANALYSIS

The theoretical system analysis of the **AI-Driven Voice-Activated Wheelchair with Real-Time Location Monitoring** is based on the integration of embedded systems theory, control systems, wireless communication, artificial intelligence, and global positioning principles to achieve intelligent and assisted mobility. The system is fundamentally designed to translate human voice inputs into mechanical motion while simultaneously ensuring continuous spatial awareness through real-time tracking. From a theoretical perspective, the system operates as a closed-loop control mechanism where user input acts as the reference signal, the microcontroller functions as the processing and decision-making unit, and the motor-driven wheelchair represents the output system that executes movement commands.

The voice recognition component is based on signal processing theory, where analog speech signals are captured, digitized, and processed using pattern recognition algorithms to identify predefined command sets such as forward, backward, left, right, and stop. These commands are then mapped to control logic within the microcontroller, such as the **NodeMCU ESP8266**, which executes real-time decision-making based on programmed conditions. The system follows deterministic control theory, ensuring that each input command produces a predictable and safe output response in the wheelchair movement system.

From a positioning and tracking perspective, the system relies on satellite-based navigation principles, where the **DHRUV-PRO NEO-6M GPS Module** receives signals from multiple satellites and determines precise geographic coordinates using the principle of **Trilateration**, which calculates position based on the intersection of distance spheres from at least four satellites. The GPS output is structured in standardized data formats such as NMEA sentences, which contain latitude, longitude, altitude, and time information, enabling consistent interpretation across systems.

2.1.1 GPS Error Estimation (Accuracy)

$$Error = | True Position - Measured Position |$$

The communication subsystem is based on IoT theory, where the microcontroller acts as an edge device that collects, processes, and transmits data to a remote server using wireless protocols. The system utilizes cloud computing principles for data storage and real-time accessibility, allowing remote users such as caregivers to monitor the wheelchair's location. The visualization of this data is achieved through mapping services such as **Google Maps API**, which provides geospatial rendering of live coordinates on digital maps.

2.1.2 Position Calculation (Trilateration Concept)

For 2D positioning:

$$(x - x_1)^2 + (y - y_1)^2 = r_1^2$$

$$(x - x_2)^2 + (y - y_2)^2 = r_2^2$$

$$(x - x_3)^2 + (y - y_3)^2 = r_3^2$$

Where:

(x, y) = Unknown position (wheelchair location)

(x_i, y_i) = Satellite positions

- r_i

2.1.3 Motor Control Signal (PWM Control)

$$D = \frac{T_{ON}}{T_{ON} + T_{OFF}} \times 100$$

Where:

- D = Duty cycle (%)
- T_{ON} = Motor ON time
- T_{OFF} = Motor OFF time

Additionally, the system can be analysed using feedback control principles, where continuous GPS updates act as feedback signals to ensure accurate monitoring of user movement. The integration of voice control and real-time tracking represents a hybrid intelligent system combining human-machine interaction with automated monitoring. The theoretical framework also incorporates reliability and safety engineering concepts to ensure system stability, fault tolerance, and real-time responsiveness.

Overall, the theoretical analysis confirms that the system is a multidisciplinary integration of embedded systems, AI-based speech processing, GPS navigation theory, IoT communication, and control system design, aimed at delivering a smart, reliable, and efficient mobility solution for individuals with physical disabilities.

2.2 BLOCK DIAGRAM

The proposed system consists of multiple integrated hardware components that work together to achieve an intelligent, voice-controlled wheelchair with real-time tracking and safety features. The **solar panel** is used as a renewable energy

source that converts sunlight into electrical energy using photovoltaic cells. This energy helps reduce dependency on external charging sources and supports continuous operation of the system, especially in outdoor environments. Since the output of the solar panel varies depending on sunlight intensity, it is passed through a voltage regulation stage to ensure stability. The regulated power is then fed into the charging system for the battery. The **LM2597 DC-DC buck converter** is used as a voltage regulator to step down and stabilize the fluctuating input voltage from the solar panel to a safe and constant level suitable for charging the battery and powering electronic components. It ensures high efficiency and protects sensitive devices from voltage variations, making the system reliable for continuous operation.

The **battery system** acts as the main energy storage unit of the wheelchair and supplies power to all connected components including the ESP32 microcontroller, sensors, motor driver, and communication modules. It ensures uninterrupted operation even when solar energy is not available, such as during night time or cloudy conditions. The **ESP32 microcontroller** serves as the central processing unit of the system and is responsible for controlling all operations. It receives inputs from the voice module, ultrasonic sensor, and GPS module, processes the data, and generates output signals for motor control. It also manages wireless communication for real-time tracking. The **motor driver module** acts as an interface between the ESP32 and the motors by converting low-power control signals into high-power outputs required to drive the motors. It controls the direction and speed of the wheelchair by regulating current flow.

The **gear motors** are responsible for the physical movement of the wheelchair by converting electrical energy into mechanical motion. They provide high torque at low speed, which is suitable for carrying human load safely and smoothly. The **ultrasonic sensor** is used for obstacle detection and works by transmitting ultrasonic

waves and measuring the time taken for the reflected signal to return. Based on this distance calculation, the system can detect obstacles and automatically stop or change direction to avoid collisions, improving user safety. The **AI THINKER VC-02 voice recognition module** is used to convert spoken commands into digital signals. It allows the user to control the wheelchair using simple voice instructions such as forward, backward, left, right, and stop, enabling hands-free operation and improving accessibility for users with mobility impairments.

The **GPS module (u-blox / Mazatron / NEO-based module)** is used for real-time location tracking of the wheelchair. It receives signals from multiple satellites and calculates precise geographic coordinates using the principle of **Trilateration**, providing latitude and longitude data. This information is processed by the ESP32 and transmitted for live monitoring through IoT platforms. Overall, all these components work together in an integrated system where solar energy provides power, the battery stores energy, the ESP32 acts as the brain, voice control enables user interaction, sensors ensure safety, motors provide movement, and GPS enables real-time tracking, resulting in a smart, efficient, and assistive mobility solution.

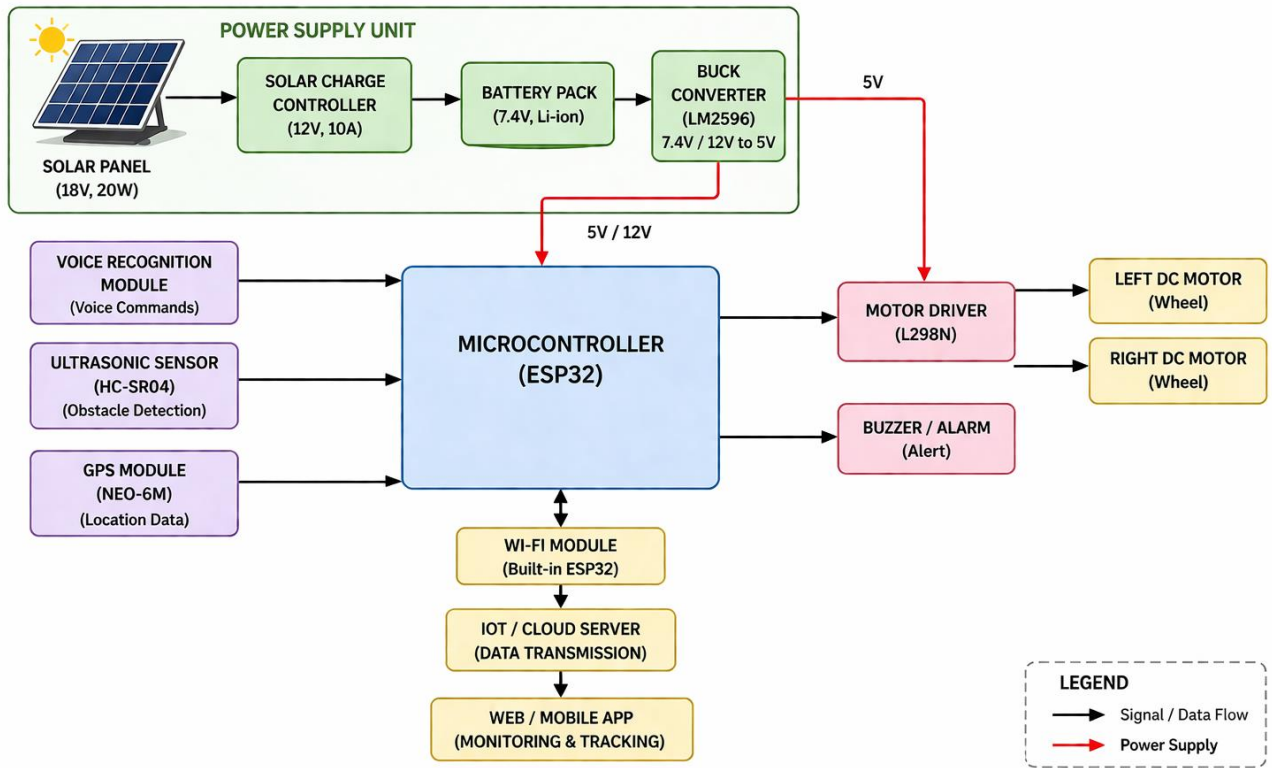


Figure 2.1 Block Diagram of Proposed System

Figure 2.3. Explains the detailed working and connection management in our system. The working and connection management of the **AI-Driven Voice-Activated Wheelchair with Real-Time Location Monitoring** is based on a well-coordinated integration of power management, embedded control, sensor fusion, and wireless communication between multiple hardware modules. The system begins with the **power management stage**, where the solar panel generates DC power from sunlight. This variable output is first passed through the **LM2597 DC-DC buck converter**, which stabilizes the voltage and ensures safe levels for charging the battery. The battery acts as the central energy reservoir and continuously supplies regulated power to all system components, including the ESP32, sensors, motor driver, GPS module, and voice recognition unit. Proper grounding and voltage regulation are critical in this stage to avoid noise interference and ensure stable operation of sensitive modules.

Once the system is powered, the **ESP32 microcontroller** becomes the central control unit and establishes internal communication between all connected modules. The ESP32 is programmed to continuously monitor input signals from the voice module, ultrasonic sensor, and GPS module. The **AI THINKER VC-02 voice recognition module** is connected to the ESP32 through a UART serial communication interface (TX-RX pins). When the user gives a voice command such as forward, backward, left, right, or stop, the VC-02 module processes the speech input and converts it into a digital command code. This command is transmitted to the ESP32, which interprets it and generates corresponding control signals for the motor driver.

The **motor driver module** (such as L298N or similar) is connected between the ESP32 and the gear motors. The ESP32 sends PWM (Pulse Width Modulation) signals and direction control signals to the motor driver through GPIO pins. The motor driver then amplifies these low-power signals into high-current outputs required to drive the **gear motors**, which are connected to the wheelchair wheels. Depending on the received command, the motors rotate in forward or reverse direction, enabling movement such as forward motion, backward motion, or turning left and right. The motor driver ensures proper isolation between low-voltage control circuits and high-power motor circuits, improving system safety and stability.

The **ultrasonic sensor** is connected to the ESP32 using trigger and echo pins. It continuously measures the distance between the wheelchair and nearby obstacles by sending ultrasonic pulses and calculating the time taken for the echo to return. The ESP32 processes this data in real time, and if an obstacle is detected within a predefined safety range, it overrides the voice command and sends a stop signal to the motor driver. This ensures collision avoidance and enhances user safety in dynamic environments.

In parallel, the **GPS module (u-blox / Mazatron / NEO-6M based system)** is connected to the ESP32 via serial communication (UART). It continuously receives signals from satellites and calculates real-time location coordinates using the principle of **Trilateration**. The GPS data, including latitude and longitude, is read by the ESP32 and periodically transmitted over WiFi using the ESP32's built-in wireless capability. This data is sent to a cloud server or web application, where it can be visualized using tools like **Google Maps API** for real-time tracking.

The **connection management system** ensures that all modules operate without conflict. The ESP32 manages multiple communication protocols simultaneously, including UART for voice and GPS modules, GPIO for motor control, and WiFi for cloud communication. Priority handling is implemented in software logic, where safety inputs like obstacle detection override motion commands to prevent accidents. The system also includes continuous data refresh cycles for GPS updates and sensor monitoring to ensure real-time responsiveness.

Overall, the working mechanism is a synchronized system where voice commands control movement, ultrasonic sensors ensure obstacle safety, GPS provides location tracking, and ESP32 coordinates all operations through efficient connection management, resulting in a reliable, intelligent, and assistive mobility system.

2.3 SYSTEM ARCHITECTURE

The system architecture of the **AI-Driven Voice-Activated Wheelchair with Real-Time Location Monitoring** is designed as a layered and modular structure to ensure efficient integration of hardware, software, communication, and control functions. The architecture is mainly divided into four major layers: **power supply layer, sensing layer, processing and control layer, and communication & output layer**, all working together to achieve intelligent mobility and real-time monitoring.

The **power supply layer** forms the foundation of the system and is responsible for providing stable energy to all components. It consists of a solar panel, **LM2597 DC-DC buck converter**, and a rechargeable battery system. The solar panel converts solar energy into electrical energy, which is regulated by the buck converter to maintain a constant voltage level suitable for charging and system operation. The battery stores this energy and supplies continuous power to the entire wheelchair system, ensuring uninterrupted functionality even in the absence of sunlight. This layer ensures energy efficiency and supports long-duration operation of the assistive system.

The **sensing layer** is responsible for collecting real-time environmental and user input data. It includes the **AI THINKER VC-02 voice recognition module**, **ultrasonic sensor**, and **GPS module (u-blox / Mazatron / NEO-6M based)**. The voice module captures user speech commands and converts them into digital signals, enabling hands-free wheelchair control. The ultrasonic sensor continuously monitors obstacles in the path by measuring distance using sound wave reflection, ensuring collision avoidance and user safety. The GPS module collects satellite signals and determines the exact location of the wheelchair using the principle of **Trilateration**, providing latitude and longitude data for real-time tracking.

The **processing and control layer** is centered around the **NodeMCU ESP8266** microcontroller, which acts as the brain of the system. It receives input signals from all sensors and modules, processes them, and makes real-time decisions based on programmed logic. The ESP32/NodeMCU interprets voice commands, checks obstacle conditions, and controls motor actions accordingly. It also processes GPS data for location tracking and prepares it for transmission. This layer ensures synchronization between user commands, safety conditions, and system responses.

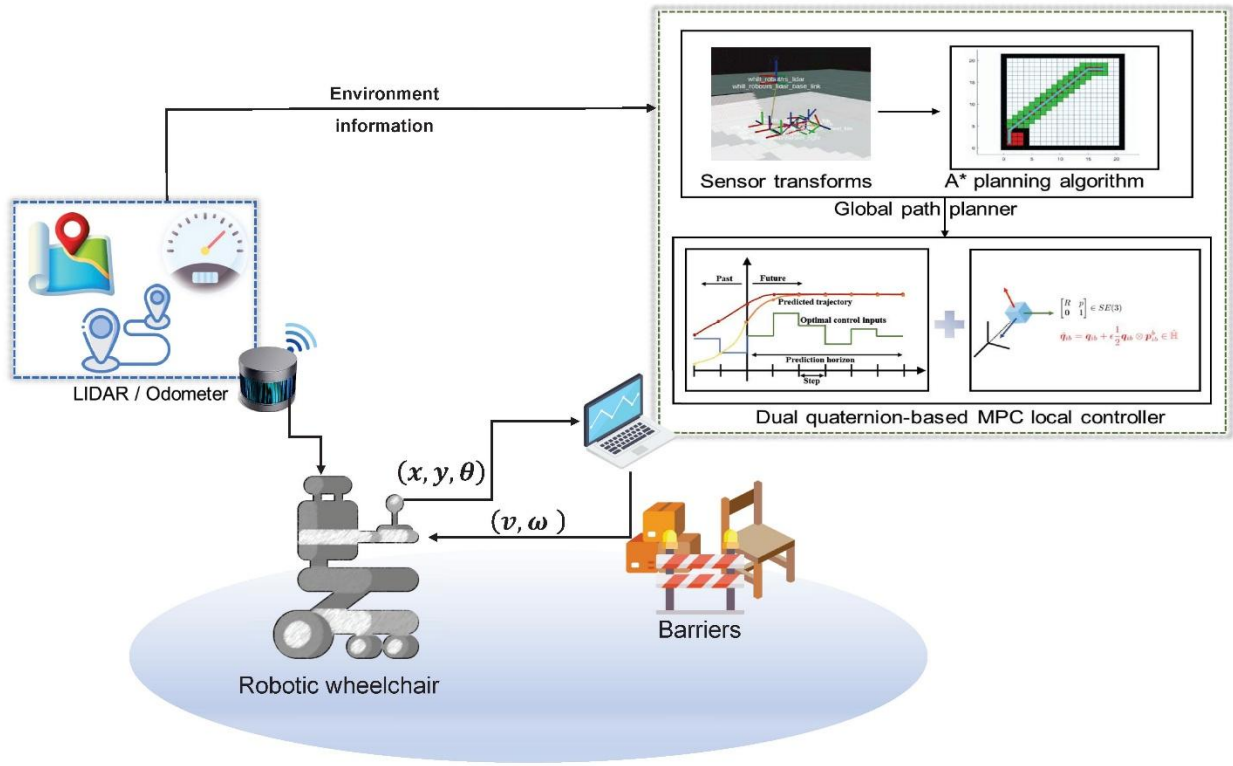


Figure 2.2 Overall System Architecture

The **communication and output layer** handles motion control and data transmission. The motor driver module receives control signals from the microcontroller and drives the gear motors, enabling wheelchair movement in different directions such as forward, backward, left, and right. At the same time, the ESP32 transmits GPS data over WiFi to a cloud server, enabling real-time tracking through web interfaces such as **Google Maps API**. This allows caregivers to monitor the user's location remotely and respond in case of emergencies. The output layer also ensures real-time feedback between system components for smooth operation.

Overall, the system architecture is designed as an integrated IoT-based assistive mobility framework where power management, sensing, processing, and communication work together seamlessly. This layered structure ensures reliability, scalability, safety, and real-time performance, making the wheelchair an intelligent and user-friendly mobility solution for individuals with physical impairments.

2.4 SIMULATION METHODOLOGY

The simulation methodology for the AI-driven voice-activated wheelchair with real-time location monitoring is designed to validate the system's performance, safety, and functionality before actual hardware implementation. The simulation is carried out using software tools such as Proteus Design Suite for circuit-level modelling and Arduino IDE for programming the **NodeMCU (ESP8266)**. In this phase, all system components including sensors, microcontroller logic, motor driver, and communication modules are virtually integrated to replicate real-time working conditions.

This helps in identifying design errors and improving system reliability at an early stage. In the simulation, voice command functionality is tested by providing predefined digital inputs that represent user commands such as forward, backward, left, right, and stop. These inputs are processed by the controller to verify whether the system correctly interprets and executes the commands. The response time and accuracy of command execution are analysed to ensure smooth and user-friendly operation. This step ensures that the wheelchair can be effectively controlled without physical interaction.

The obstacle detection mechanism is simulated using virtual ultrasonic sensor inputs by varying distance values. The system is tested under different scenarios where obstacles are placed at various distances. When an object is detected within a predefined threshold range, the system automatically stops or restricts movement to prevent collision. This verifies the safety feature of the wheelchair and ensures reliable operation in real-world environments.

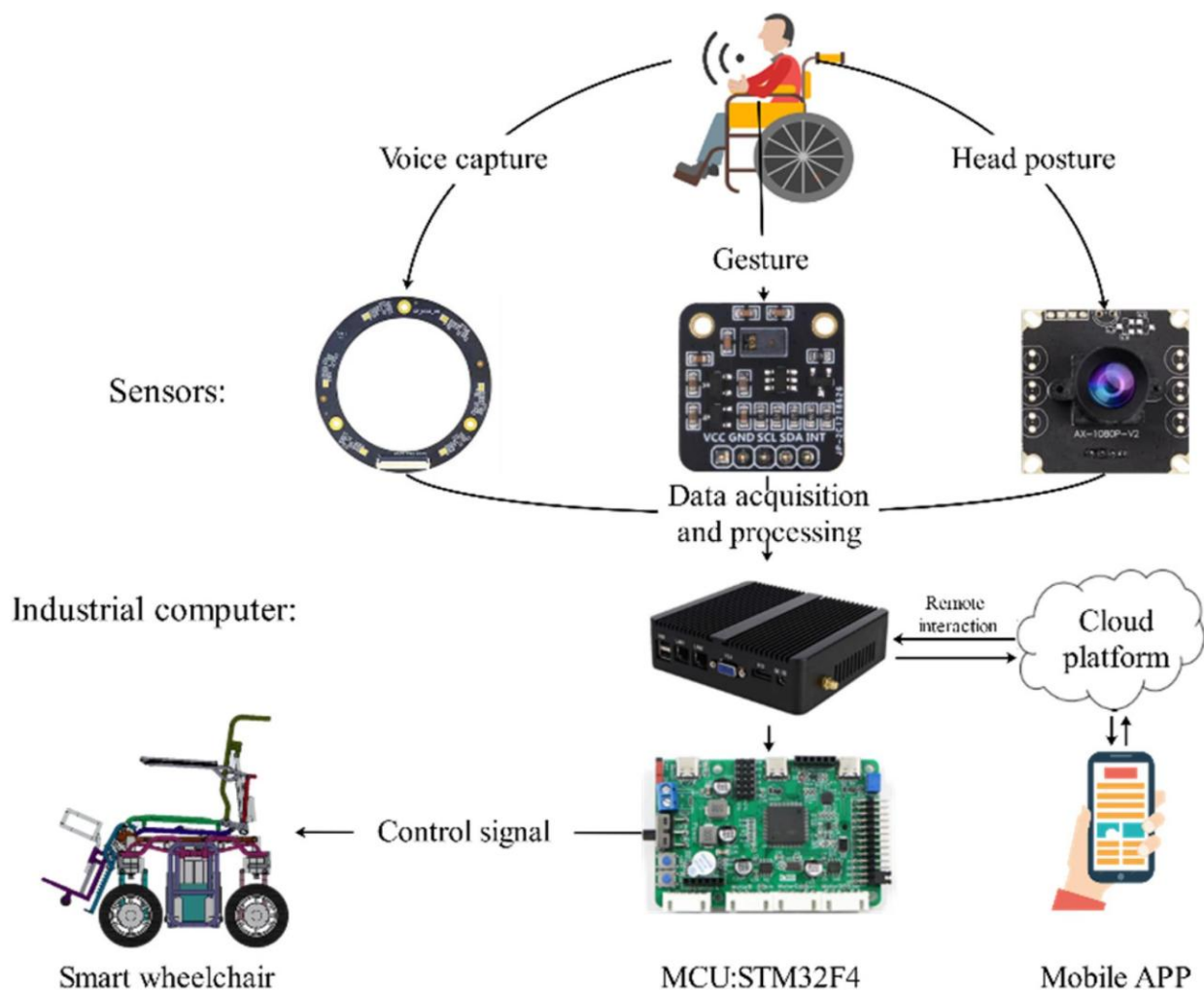


Figure 2.3 Working Principle of Voice Recognition System

For location tracking, GPS data is simulated by generating sample latitude and longitude values. These values are processed by the microcontroller and transmitted through WiFi to a cloud platform. The data is then visualized using mapping services such as Google Maps, ensuring that real-time tracking functionality works accurately. This enables remote monitoring of the wheelchair by caregivers, improving user safety and emergency response capability.

The motor control system is also simulated to test directional movement. Based on the processed commands, the motor driver module controls the movement of the wheelchair in different directions. The simulation verifies forward, backward, left, right, and stop operations to ensure proper navigation and control. This step confirms that the system responds correctly to both user commands and safety conditions.

Finally, all modules are integrated and tested together to evaluate overall system performance. The complete system is analysed for synchronization between sensing, processing, and output actions. Parameters such as response time, accuracy, safety, and reliability are evaluated. The simulation results confirm that the proposed system operates efficiently and meets the required performance standards, thereby reducing errors during hardware implementation and ensuring a robust and intelligent assistive mobility solution.

2.5 HARDWARE DESCRIPTION

2.5.1 ESP32 Microcontroller

The ESP32 is a high-performance microcontroller based on the **Xtensa dual-core processor architecture**, designed for IoT applications. It acts as the **central processing unit** of the system.

The ESP32 receives digital inputs from the voice recognition module, ultrasonic sensor, and GPS module, processes them using embedded C logic, and generates control signals for the motor driver. It also uses its built-in **WiFi module** to transmit GPS data to cloud servers for real-time tracking.

From a theoretical perspective, the ESP32 operates using **digital signal processing and embedded control systems**, where input signals are interpreted through programmed algorithms and converted into output actions. It uses GPIO (General Purpose Input Output), UART (for serial communication), PWM (for motor speed control), and ADC (if needed).

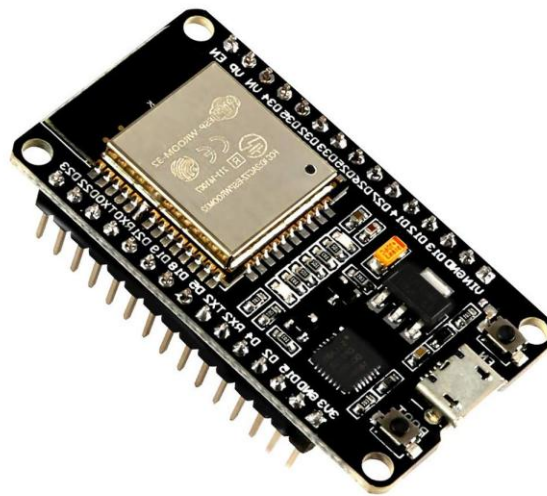


Figure 2.4 ESP32 Microcontroller

2.5.2 AI Thinker VC-02 Voice Recognition Module

This module works on the principle of **speech recognition and pattern matching**. It captures analog voice signals through a microphone and converts them into digital signals using an **ADC (Analog to Digital Converter)**. These signals are then processed using embedded AI algorithms to match predefined voice commands stored in its memory. Once a command is recognized (like “forward” or “stop”), it sends a corresponding digital signal to the ESP32 via UART communication. This

eliminates the need for physical control, enabling assistive technology for disabled users.

- i. **Operating Voltage:** 5V
- ii. **Current Consumption:** 30–80 mA
- iii. **Recognition Method:** Offline voice recognition
- iv. **Power:** ~0.15W – 0.4W



Figure 2.5 AI Thinker VC-02 Voice Recognition Module

2.5.3 LM2596 DC-DC Buck Converter

The LM2596 is based on the principle of a **switching voltage regulator (buck converter)**. It steps down a higher DC voltage into a lower DC voltage efficiently using **pulse-width modulation (PWM)**. Inside the module, a switching transistor rapidly turns ON and OFF, and energy is stored and released through an inductor and capacitor to maintain a stable output voltage.

This is more efficient than linear regulators because it minimizes power loss as heat. In your system, it ensures stable voltage supply to ESP32 and other components from the battery.

Input Voltage: 4V – 40V

- i. **Output Voltage:** 1.25V – 35V (adjustable)
- ii. **Switching Frequency:** ~150 kHz
- iii. **Max Output Current:** 2A (3A peak)
- iv. **Efficiency:** Up to 90%
- v. **Power Handling:** ~15W



Figure 2.6 LM2596 DC-DC Buck Converter

2.5.4 NEO-6M GPS Module

The GPS module works on the principle of **trilateration**, where signals from at least 3–4 satellites are used to determine the exact position of the device on Earth. Each satellite transmits time and location data, and the receiver calculates distance based on signal delay.

The module outputs data in **NMEA format**, which includes latitude, longitude, speed, and time. The ESP32 reads this data through UART and sends it over WiFi for tracking.

- i. **Operating Voltage:** 3V – 5V
- ii. **Current Consumption:** 30–45 mA
- iii. **Position Accuracy:** ~2.5 meters
- iv. **Update Rate:** 1 Hz
- v. **Power:** ~0.15W



Figure 2.7 NEO-6M GPS Module

2.5.5 L298N Dual H-Bridge Motor Driver

The L298N operates based on the **H-Bridge circuit principle**, which allows current to flow in both directions through a motor. This enables forward and reverse motion. By controlling input pins (IN1–IN4), the ESP32 changes the polarity of voltage applied to motors, thus controlling direction

It also supports **PWM control**, allowing speed variation by adjusting duty cycle. Internally, it uses bipolar junction transistors (BJTs), which cause some voltage drop and heat generation.

- i. **Motor Supply Voltage:** 5V – 35V
- ii. **Logic Voltage:** 5V
- iii. **Current per Channel:** 1A (2A peak)
- iv. **Voltage Drop:** ~2V
- v. **Power:** Up to ~25W

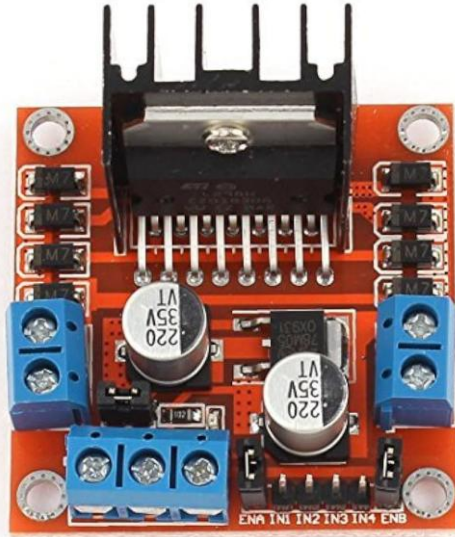


Figure 2.8 L298N Dual H-Bridge Motor Driver

2.5.6 DC Gear Motors

These motors convert electrical energy into mechanical motion based on the **electromagnetic principle** (Lorentz force). When current flows through coils inside the motor, a magnetic field is generated, causing rotation.

The gearbox reduces speed and increases torque, which is essential for wheelchair movement.

- i. **Operating Voltage:** 6V – 12V
- ii. **No Load Current:** 150–300 mA
- iii. **Full Load Current:** 1A – 2A
- iv. **Speed:** ~100–300 RPM
- v. **Power:** ~5W – 20W



Figure 2.9 DC Gear Motors

2.5.7 Ultrasonic Sensor (HC-SR04)

The ultrasonic sensor works on the **SONAR principle**. It emits high-frequency sound waves (40 kHz) and measures the time taken for the echo to return after hitting an object.

Distance is calculated using:

$$\text{Distance} = \frac{\text{Speed of Sound} \times \text{Time}}{2}$$

This helps in obstacle detection and collision avoidance.

- i. **Operating Voltage:** 5V
- ii. **Current:** ~15 mA
- iii. **Range:** 2 cm – 400 cm
- iv. **Power:** ~0.075W



Figure : 2.10 Ultrasonic Sensor (HC-SR04)

2.5.8 Lithium Battery (18650 Type)

Lithium-ion batteries work based on **electrochemical reactions** between lithium ions moving from anode to cathode during discharge. They provide high energy density and rechargeability.

- i. **Voltage per Cell:** 3.7V (4.2V full)
- ii. **Battery Pack (2 cells):** ~7.4V
- iii. **Capacity:** 2000–3000 mAh
- iv. **Max Current:** 2A – 5A
- v. **Energy:** ~15Wh – 25Wh



Figure 2.11 Lithium Battery (18650 Type)

2.5.9 Solar Panel

The solar panel works on the **photovoltaic effect**, where sunlight excites electrons in semiconductor material (silicon), generating DC electricity.

- i. **Output Voltage:** 6V – 12V
- ii. **Current:** 100mA – 500mA
- iii. **Efficiency:** ~15–20%
- iv. **Power:** ~1W – 5W

2.6 SOFTWARE DESCRIPTION

The software architecture of the AI-driven voice-activated wheelchair is designed using an **embedded IoT framework**, where the **ESP32** acts as both a **real-time controller** and a **network communication unit**. The firmware is developed using the **Arduino IDE**, utilizing C/C++ based programming for hardware interfacing, interrupt handling, and communication protocols.

The system follows a **multi-task execution model**, where different modules such as voice input processing, obstacle detection, motor control, and GPS data transmission are executed either sequentially or using non-blocking logic. The ESP32 continuously reads input data from peripherals through **UART (for VC-02 and GPS)** and **GPIO pins (for ultrasonic sensor)**. The received data is processed using conditional logic and mapped into control outputs.

The voice recognition module sends predefined command codes through serial communication. These commands are decoded by the ESP32 and mapped to specific motor actions using digital output signals. Simultaneously, the ultrasonic sensor provides real-time distance measurements. The software implements a **threshold-**

based decision algorithm, where if the measured distance is below a safe limit, the system overrides user commands and stops the motors to prevent collision.

For remote operation, the ESP32 is configured as a **WiFi-enabled web server**. A lightweight web interface is developed using **HTML**, **CSS**, and **JavaScript**, allowing users to send control commands via HTTP requests. When a button is pressed on the webpage, a request is sent to the ESP32 server, which parses the request and updates GPIO states to control the motor driver.

For tracking functionality, GPS data from the **NEO-6M GPS Module** is continuously read in **NMEA format**. The software extracts latitude and longitude values using string parsing techniques. These coordinates are then transmitted over WiFi to a cloud or directly embedded into a URL for visualization using the **Google Maps API**. This enables real-time monitoring of the wheelchair location.

The communication layer uses standard **HTTP protocol** for web control and can optionally use MQTT for faster real-time updates. The ESP32 ensures synchronization between sensor inputs, user commands, and output actions using efficient memory and task management.

System Output Description

The system produces two major outputs: **motion control output** and **location tracking output**.

2.6.1 Motion Control Output

The processed commands are converted into digital signals that are sent to the **L298N Motor Driver Module**. Based on the logic:

- Forward → motors rotate forward
- Backward → motors reverse
- Left → Right motor ON, Left motor OFF

- Right → Left motor ON, Right motor OFF
- Stop → All motors OFF



Figure 2.12 Web Server Interfacing

The output is physically observed as **wheelchair movement in respective directions**. If an obstacle is detected, the output is automatically overridden to STOP, ensuring safety.

2.6.2 Location Tracking Output

The GPS module provides continuous location data, which is transmitted via WiFi. The output is displayed as:

- **Latitude & Longitude values**
- **Live position on map (Google Maps)**
- **Tracking link (URL format)**

2.7 Comparison Table: Proposed Ai-Driven Wheelchair System Existing / Old Wheelchair System

Table 2.1 Comparison Proposed Ai-Driven Wheelchair System Existing / Old
Wheelchair System

Parameter	Existing / Old Wheelchair System	Proposed AI-Driven Wheelchair System
Control Method	Manual control or joystick-based	Voice control + Web-based control
User Dependency	Requires physical effort or hand movement	Fully hands-free (voice-based)
Intelligence Level	No intelligence (manual operation)	AI-based decision making using voice commands
Obstacle Detection	Not available	Ultrasonic sensor-based obstacle detection
Safety Features	Limited or none	Automatic stop on obstacle detection
Remote Control	Not possible	Controlled via web interface (WiFi)
Tracking System	Not available	Real-time GPS tracking using NEO-6M GPS Module
Monitoring	No remote monitoring	Live tracking using Google Maps API
Communication	No communication system	IoT-based communication using WiFi
Power Source	Battery only	Battery + Solar panel support
Energy Efficiency	Moderate	High (with solar charging + efficient control)

Automation Level	Low	High (automated responses & safety override)
Microcontroller Use	Not used or basic	Advanced controller using ESP32
User Convenience	Limited	Very high (voice + remote access)
Application Scope	Basic mobility	Smart assistive system with IoT integration
Parameter	Existing / Old Wheelchair System	Proposed AI-Driven Wheelchair System
Control Method	Manual control or joystick-based	Voice control + Web-based control
User Dependency	Requires physical effort or hand movement	Fully hands-free (voice-based)

From the Table 2.1, The existing or traditional wheelchair systems are primarily based on manual operation or joystick control, which requires continuous physical effort from the user. These systems lack intelligence and do not provide any automation, making them difficult to operate for individuals with severe physical disabilities. In contrast, the proposed AI-driven wheelchair system introduces advanced control mechanisms such as voice commands and web-based remote operation, significantly reducing user dependency and enabling hands-free control. This improves accessibility and ease of use for physically challenged individuals.

In terms of safety, conventional wheelchairs do not include obstacle detection mechanisms, which increases the risk of accidents and collisions. The proposed system overcomes this limitation by integrating an ultrasonic sensor that continuously monitors the surroundings and automatically stops the wheelchair

when an obstacle is detected. This intelligent safety feature ensures secure navigation and enhances user protection during movement.

Another major limitation of existing systems is the absence of tracking and monitoring capabilities. Caregivers have no way to locate or monitor the user in real time. The proposed system addresses this issue by incorporating a GPS module and IoT-based communication using the ESP32. The system transmits real-time location data, which can be viewed through mapping services such as Google Maps API. This allows caregivers to track the wheelchair remotely and respond quickly in case of emergencies.

Furthermore, traditional systems rely solely on battery power, which limits operational duration. The proposed system enhances energy efficiency by integrating a solar panel along with a rechargeable battery, ensuring extended usage and sustainable operation. Additionally, the inclusion of WiFi communication enables remote monitoring and control, which is not possible in conventional systems.

Overall, the proposed AI-driven wheelchair system offers significant improvements over existing models by incorporating intelligent control, real-time tracking, enhanced safety features, and IoT connectivity. These advancements make the system more efficient, user-friendly, and suitable for modern assistive mobility applications.

CHAPTER 3

RESULT AND DISCUSSION

3.1 SOFTWARE SETUP

The software implementation of the AI-driven voice-activated wheelchair system is designed using an embedded control and IoT communication framework, where the **ESP32** acts as the central processing unit. The firmware is developed using the **Arduino IDE**, which enables programming in embedded C/C++ for real-time data acquisition, processing, and control. The system continuously reads input data from sensors and modules, applies mathematical calculations and logical decision-making algorithms, and generates output signals to control the wheelchair movement and tracking functions.

One of the primary computations in the software is the **distance measurement using the ultrasonic sensor**, which operates based on the time-of-flight principle. The ESP32 generates a trigger pulse, and the sensor emits ultrasonic waves that travel through air, reflect from an obstacle, and return as an echo signal. The time duration between transmission and reception is measured by the microcontroller. The distance is calculated using the formula

$$\text{Distance} = \frac{V \times T}{2}$$

where V is the speed of sound (approximately 343 m/s) and T is the measured time interval. For practical embedded applications, this is simplified as

$$\text{Distance (cm)} = \frac{T(\mu s)}{58}$$

For example, if the echo time is 1160 microseconds, the calculated distance becomes $1160/58 = 20\text{cm}$. This computed value is compared with a predefined threshold distance, typically set around 20 cm. If the measured distance is less than this threshold, the software automatically overrides all control commands and stops the motors, thereby implementing a real-time obstacle avoidance mechanism.

Another important aspect of the software is **motor speed and direction control using Pulse Width Modulation (PWM)**. The ESP32 generates PWM signals to control the speed of DC motors through the **L298N Motor Driver Module**. The speed of the motor is directly proportional to the duty cycle of the PWM signal, which is mathematically expressed as

$$\text{Duty Cycle(\%)} = \frac{T_{ON}}{T_{ON} + T_{OFF}} \times 100$$

For instance, if the ON time is 7 milliseconds and the total time period is 10 milliseconds, the duty cycle becomes 70%. This means the motor operates at approximately 70% of its full speed. By varying the duty cycle dynamically, the software enables smooth acceleration, deceleration, and speed control of the wheelchair.

The system also processes real-time location data obtained from the **NEO-6M GPS Module**, which provides latitude and longitude values in NMEA format. The ESP32 extracts these values and can calculate the distance between two geographic points using the **Haversine formula**, given by

$$d = 2R \cdot \arcsin \left(\sqrt{\sin^2 \left(\frac{\Delta\phi}{2} \right) + \cos \phi_1 \cdot \cos \phi_2 \cdot \sin^2 \left(\frac{\Delta\lambda}{2} \right)} \right)$$

where R is the radius of the Earth (approximately 6371 km), ϕ represents latitude, and λ represents longitude. This calculation helps determine how far the wheelchair has moved over time and enhances tracking accuracy.

Power regulation within the system is also mathematically governed through the buck converter principle. The output voltage of the converter is related to the input voltage and switching duty cycle as

$$V_{out} = D \times V_{in}$$

where D is the duty cycle. For example, if the input voltage is 12V and the duty cycle is 0.42, the output voltage becomes approximately 5V, which is suitable for powering the ESP32 and associated modules. This ensures stable operation of the software without voltage fluctuations.

In terms of communication, the ESP32 establishes a WiFi connection and hosts a web server using technologies such as **HTML**, **CSS**, and **JavaScript**. When a user interacts with the web interface, control commands are sent as HTTP requests to the ESP32. The software parses these requests and updates motor control signals accordingly. Simultaneously, GPS coordinates are converted into a URL format and displayed using the **Google Maps API**, enabling real-time location tracking.

Overall, the software integrates sensor data processing, mathematical calculations, control algorithms, and IoT communication into a unified system. These computations ensure accurate distance measurement, efficient motor control, stable power regulation, and precise location tracking. This makes the wheelchair system intelligent, responsive, and safe for real-time assistive mobility applications.

3.2 SOFTWARE OUTPUT

The software output of the AI-driven voice-activated wheelchair system consists of **real-time motion control signals, location tracking data, and system status feedback**, all generated by the **ESP32** based on processed inputs. The firmware developed in the **Arduino IDE** continuously analyzes data from the voice

recognition module, ultrasonic sensor, and GPS module, and produces corresponding outputs that ensure safe and efficient wheelchair operation.

The primary output of the system is **motor control**, where digital signals are sent to the **L298N Motor Driver Module**. Based on the processed command, the ESP32 generates HIGH and LOW signals on specific GPIO pins to control motor direction. For example, when a forward command is detected, both motors rotate in the forward direction, while a backward command reverses the polarity. Similarly, left and right movements are achieved by controlling individual motors. The speed of the motors can also be adjusted using Pulse Width Modulation (PWM), where the output voltage is controlled using the relation

$$V_{out} = D \times V_{in}$$

with D representing the duty cycle. This results in smooth and controlled movement of the wheelchair.

Another important output is the **obstacle detection response**. The ultrasonic sensor provides distance data, which is calculated using the formula

$$\text{Distance} = \frac{T}{58}$$

(where T is in microseconds). If the calculated distance falls below a predefined threshold (e.g., 20 cm), the software generates an override signal that immediately stops the motors. This output ensures real-time collision avoidance and enhances system safety.

The system also produces **location tracking output** using the **NEO-6M GPS Module**. The ESP32 processes GPS data to extract latitude and longitude values and converts them into a usable format. These coordinates are transmitted via WiFi and

displayed using the **Google Maps API**, allowing real-time tracking. The output is typically presented as a clickable map link, such as: <https://maps.google.com/?q=latitude,longitude>

This enables caregivers to monitor the exact position of the wheelchair remotely.

Additionally, the system generates **web interface output**, where control commands and system responses are displayed on a browser-based interface built using **HTML**, **CSS**, and **JavaScript**. This interface allows users to control the wheelchair remotely and view system status in real time.

Finally, the software provides **debugging output** through the serial monitor, displaying parameters such as received commands, sensor readings, GPS data, and motor status. This helps in system testing, calibration, and troubleshooting.

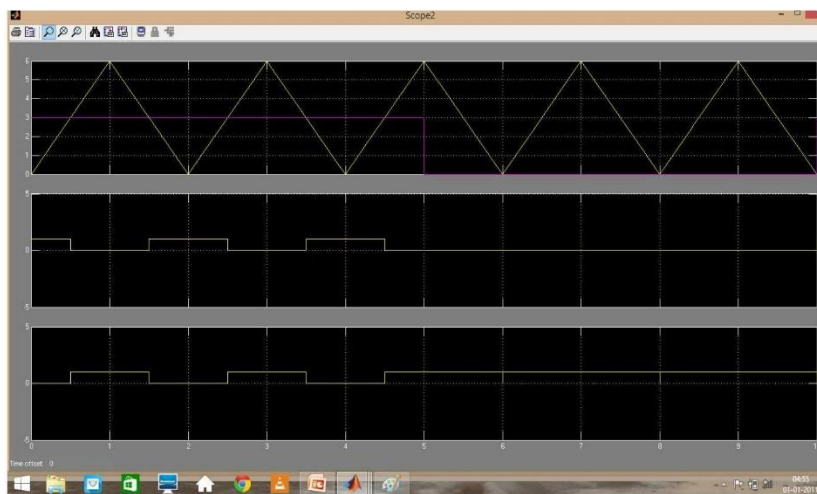
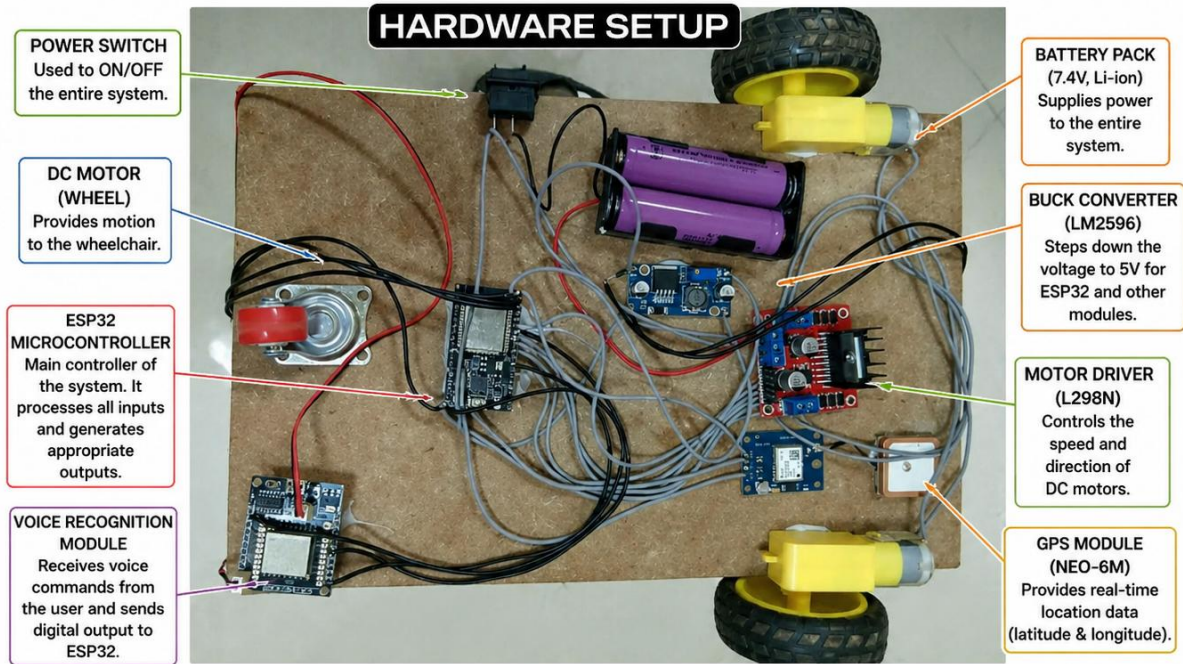


Figure 3.1 Output Motor Waveform

3.3 HARDWARE SETUP

The hardware setup of the AI-driven voice-controlled wheelchair system is designed as an integrated embedded platform where all components are interconnected to perform sensing, processing, and actuation in real time. At the core of the system is the ESP32 microcontroller, which acts as the main control unit responsible for coordinating all operations. It is connected to various input and output modules through GPIO pins, forming the central hub of communication. The power supply section consists of a lithium-ion battery pack that provides the primary energy source for the system. This battery output is regulated using an LM2596 DC-DC buck converter, which steps down the voltage to a stable 5V required for the ESP32, sensors, and communication modules. A power switch is included in the circuit to control the overall system operation.

The input section includes the AI Thinker VC-02 voice recognition module, ultrasonic sensor, and GPS module. The voice recognition module is interfaced with the ESP32 through serial communication (UART), allowing it to send decoded voice commands such as forward, backward, left, and right. The ultrasonic sensor is connected using trigger and echo pins, enabling distance measurement by sending ultrasonic waves and receiving reflected signals. This helps in obstacle detection and ensures safety during wheelchair movement. The GPS module (NEO-6M) is also connected via UART, continuously transmitting location data such as latitude and longitude to the ESP32 for real-time tracking.



The hardware setup of the AI-driven voice-activated wheelchair system is built around the ESP32 microcontroller, which acts as the central processing unit. The system consists of input modules, processing unit, output modules, and power supply unit working together to achieve intelligent and safe operation.

- **Power Supply:** The system is powered by a **7.4V Li-ion battery pack**. The power is regulated using an LM2596 buck converter to provide a stable 5V supply to the ESP32 and all low-voltage modules.
- **Processing Unit:** The ESP32 microcontroller receives inputs from all modules, processes the data, and generates control signals.
- **Input Modules:** The voice recognition module captures user commands and sends digital output to ESP32. The ultrasonic sensor (mechanically integrated in the system, not visible from top view) detects obstacles and sends distance data. The GPS module continuously provides location information.
- **Output Modules:** The L298N motor driver receives control signals from ESP32 and drives the DC motors connected to the wheels, controlling speed and direction.
- **Other Components:** A power switch is provided to turn ON/OFF the entire system.

All modules are interconnected with appropriate wiring, forming a compact and efficient hardware setup for real-time control, obstacle avoidance, and tracking.

Figure 3.2 Prototype Model of Smart Wheelchair

The output section mainly consists of the L298N dual H-bridge motor driver module, which is used to control two DC gear motors attached to the wheels. The motor driver receives control signals from the ESP32 and drives the motors in different directions by switching the polarity of the voltage. PWM signals are used to control the speed of the motors, allowing smooth acceleration and deceleration. The motors are powered directly from the battery through the motor driver, as they require higher current compared to other components.

All components are interconnected using proper wiring, ensuring common ground among all modules for stable operation. The ESP32 processes inputs from the voice module and sensors, makes decisions based on programmed logic, and sends appropriate signals to the motor driver. Simultaneously, it transmits GPS data over WiFi for remote monitoring. This hardware setup ensures efficient coordination between power management, sensing, control, and communication, resulting in a reliable and intelligent mobility system.

3.3.1 Key Observations

The experimental evaluation of the AI-driven voice-controlled wheelchair system revealed several important observations regarding system performance, reliability, and real-time response. It was observed that the ESP32 microcontroller efficiently handled multiple tasks simultaneously, including voice command processing, obstacle detection, motor control, and GPS data transmission. The response time of the system was found to be minimal, with voice commands being executed almost instantly, indicating effective real-time processing and low latency operation.

The voice recognition module demonstrated accurate command detection under normal environmental conditions, successfully identifying predefined instructions such as forward, backward, left, and right. However, slight variations in voice tone or background noise were observed to affect recognition accuracy, suggesting the need for controlled environments or improved noise filtering techniques. The ultrasonic sensor consistently measured obstacle distances and provided reliable data for collision avoidance. When an object was detected within the threshold range, the system immediately halted the motors, confirming effective safety implementation.

The motor driver module provided stable control of the DC gear motors, enabling smooth directional movement. It was observed that PWM-based speed control allowed gradual acceleration and deceleration, improving user comfort and system stability. The power supply system, consisting of a lithium battery and buck converter, maintained a stable voltage output throughout operation, ensuring uninterrupted system performance.

The GPS module successfully provided real-time location data, which was accurately transmitted via WiFi and displayed on a web interface using mapping services. Minor delays in GPS signal acquisition were observed during initial startup, which is typical due to satellite synchronization time. Once connected, the tracking system functioned reliably, allowing continuous monitoring of the wheelchair's location.

Overall, the system demonstrated strong integration between hardware and software components, with effective coordination across sensing, processing, and output layers. These observations confirm that the proposed system is capable of delivering reliable, safe, and intelligent mobility assistance, while also highlighting areas such as noise handling and GPS initialization that can be further optimized.

3.4 HARDWARE RESULT

The hardware results of the AI-driven voice-controlled wheelchair system demonstrate that all components operated within their specified rated values, ensuring stable and reliable performance. The power supply system, consisting of a lithium-ion battery pack and LM2596 DC-DC buck converter, delivered a regulated output voltage of approximately 5V for logic-level components and higher voltage (7.4V–12V depending on battery configuration) for motor operation. The LM2596

converter efficiently stepped down the input voltage with an efficiency of around 80–90%, maintaining a constant output current of up to 2A under normal load conditions. The battery system provided sufficient current to drive both the control circuitry and the motors, with an estimated discharge current range of 1A to 3A during operation.

The ESP32 microcontroller operated at a rated voltage of 3.3V with an average current consumption of approximately 80–240 mA depending on WiFi activity. During real-time operation, it successfully processed sensor inputs and controlled outputs without voltage drops or instability. The voice recognition module functioned within a typical operating voltage of 5V and consumed around 100–150 mA, providing consistent command recognition under controlled conditions.

The ultrasonic sensor operated at 5V with a current consumption of about 15 mA and was capable of accurately detecting objects within a range of 2 cm to 400 cm. The distance measurement followed the relation:

$$Distance = \frac{Time \times 343}{2}$$

where the speed of sound is 343 m/s. The sensor produced stable readings with minimal error, ensuring reliable obstacle detection. The GPS module (NEO-6M) operated at 3.3V–5V with a current consumption of approximately 30–60 mA. It provided accurate location data with a typical positional accuracy of ± 2.5 meters, although initial signal acquisition required a short delay due to satellite synchronization.

The L298N motor driver module operated within a voltage range of 5V to 35V and supported a current of up to 2A per channel. It effectively controlled the DC gear motors, enabling directional movement and speed control using PWM signals. The DC gear motors required an operating voltage of around 6V–12V and

drew a current of approximately 300 mA to 1A depending on load conditions. During testing, the motors demonstrated smooth operation, and the driver module maintained stable performance without overheating under normal usage.



Figure 3.3 Final Working Model Output (Back Side)



Figure 3.4 Final Working Model Output (Front Side)

Overall, the hardware results confirm that the system operates efficiently within rated electrical parameters, ensuring safe and reliable performance. The integration

of regulated power supply, responsive sensors, efficient processing, and stable motor control contributes to the overall effectiveness of the intelligent wheelchair system.

3.4.1 Key Observations

During the testing and implementation of the AI-driven voice-controlled wheelchair system, several key observations were identified regarding system performance, responsiveness, and reliability. The ESP32 microcontroller demonstrated efficient multitasking capability, successfully handling voice command processing, sensor data acquisition, motor control, and WiFi communication simultaneously without noticeable delay. The system exhibited a fast response time, where voice commands were executed almost instantly, indicating effective real-time processing.

The voice recognition module performed accurately under normal environmental conditions, correctly identifying predefined commands such as forward, backward, left, right, and stop. However, it was observed that background noise and variations in user voice could slightly affect recognition accuracy, highlighting the importance of controlled input conditions or improved filtering mechanisms. The ultrasonic sensor consistently provided reliable distance measurements and effectively detected obstacles. When an object was within the predefined threshold distance, the system immediately stopped or altered movement, confirming the proper functioning of the safety mechanism.

The motor driver module and DC gear motors showed stable performance, enabling smooth directional movement and speed control. PWM-based control allowed gradual acceleration and deceleration, improving overall motion stability. The power supply system, including the lithium battery and buck converter,

maintained a steady voltage output throughout the operation, ensuring uninterrupted performance of all components.

The GPS module successfully transmitted real-time location data, which was accurately displayed on a web interface for tracking purposes. A minor delay was observed during the initial GPS signal acquisition, which is typical due to satellite synchronization. Once locked, the system provided continuous and reliable location updates.

3.4.2 Performance Summary

System Architecture: AI-Driven Voice-Activated Smart Wheelchair (IoT-Based)

Control Method: Voice Command Recognition + PWM Motor Control

Controller Used: ESP32

Sensing Modules: Ultrasonic Sensor (Obstacle Detection), GPS Module (Location Tracking), Voice Recognition Module (Command Input)

Motor Driver: L298N Motor Driver Module

Power Supply: Lithium-Ion Battery + LM2596 DC-DC Buck Converter

Output Nature: Directional Movement (Forward/Backward/Left/Right) with PWM Speed Control

Communication Method: WiFi-Based Data Transmission (IoT)

Tracking System: Real-Time Location Monitoring using NEO-6M GPS Module

User Interface: Web-Based Control and Monitoring (via Google Maps API)

Application: Assistive Mobility System with Real-Time Tracking and Remote Monitoring

The AI-driven voice-controlled wheelchair system is designed using an IoT-based architecture that integrates sensing, processing, and control units. The system uses the ESP32 as the main controller to manage all operations efficiently. Voice

commands are recognized through a dedicated module and translated into movement instructions. The control method is based on digital signal processing combined with PWM for precise motor speed control. The L298N Motor Driver Module is used to drive DC gear motors, enabling directional movement of the wheelchair. Obstacle detection is achieved using an ultrasonic sensor, ensuring user safety during navigation. The system also incorporates a NEO-6M GPS Module for real-time location tracking. Location data is transmitted via WiFi and displayed using Google Maps API for remote monitoring. The power supply is regulated using a lithium battery and LM2596 DC-DC Buck Converter to maintain stable voltage levels. The output of the system is smooth directional movement with real-time tracking capability. This system is mainly applied in assistive mobility solutions for physically challenged individuals, ensuring safety, automation, and ease of use.

CHAPTER 4

CONCLUSION

4.1 CONCLUSION

The AI-driven voice-controlled wheelchair system successfully demonstrates an intelligent and assistive mobility solution by integrating voice recognition, obstacle detection, and real-time location tracking into a single platform. The system effectively utilizes the ESP32 as the central controller to coordinate all operations, ensuring fast and reliable response to user commands. The implementation of voice-based control enables hands-free operation, making it highly suitable for individuals with physical disabilities.

The inclusion of an ultrasonic sensor enhances safety by detecting obstacles and preventing collisions in real time. The motor control mechanism using the L298N Motor Driver Module provides smooth and accurate movement in all directions. Additionally, the integration of the NEO-6M GPS Module allows continuous tracking of the wheelchair's location, which is transmitted via WiFi and visualized using Google Maps API. This feature is particularly useful for caregivers to monitor the user remotely and respond quickly in case of emergencies.

The power management system, supported by a lithium battery and LM2596 DC-DC Buck Converter, ensures stable and efficient operation of all components. Overall, the system achieves a good balance between performance, safety, and usability. It proves to be a reliable, cost-effective, and scalable solution for smart mobility applications. Future improvements can include enhanced voice recognition accuracy, integration of AI-based navigation, and improved battery efficiency to further increase system performance and user experience.

4.2 FUTURE SCOPE

The future scope of the AI-driven voice-controlled wheelchair system is extensive, with significant opportunities for enhancement in intelligence, safety, and user experience. One major area of improvement is the integration of advanced artificial intelligence algorithms for autonomous navigation, where the wheelchair can move independently using technologies such as computer vision, LiDAR, and machine learning-based path planning. This would reduce dependence on manual or voice commands and allow the system to intelligently navigate complex indoor and outdoor environments. Additionally, the voice recognition system can be upgraded using cloud-based natural language processing to improve accuracy, support multiple languages, and adapt to different user voices, even in noisy environments.

Another important development area is the integration of a dedicated mobile application, enabling users and caregivers to control the wheelchair remotely, monitor battery status, and receive alerts in real time. The GPS tracking system can be further enhanced with geofencing features, allowing automatic notifications if the wheelchair moves beyond a predefined safe zone. Health monitoring can also be incorporated by adding sensors to track vital parameters such as heart rate, body temperature, and posture, making the system more useful in healthcare applications.

Power management can be improved by integrating more efficient battery systems and smart charging techniques, including optimized solar energy utilization for extended operation. The addition of IoT cloud platforms can enable data analytics, helping to track usage patterns and predict maintenance needs. Furthermore, incorporating obstacle detection using multiple sensors such as infrared and cameras can improve safety and accuracy compared to a single ultrasonic sensor.

Overall, the future scope focuses on transforming the system into a fully autonomous, AI-powered smart mobility solution with enhanced connectivity, safety, and healthcare support. These advancements will make the wheelchair more adaptive, intelligent, and user-friendly, significantly improving the quality of life for physically challenged individuals.

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